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Development and application of a comprehensive numerical simulation model for microalgal pneumatic photobioreactors

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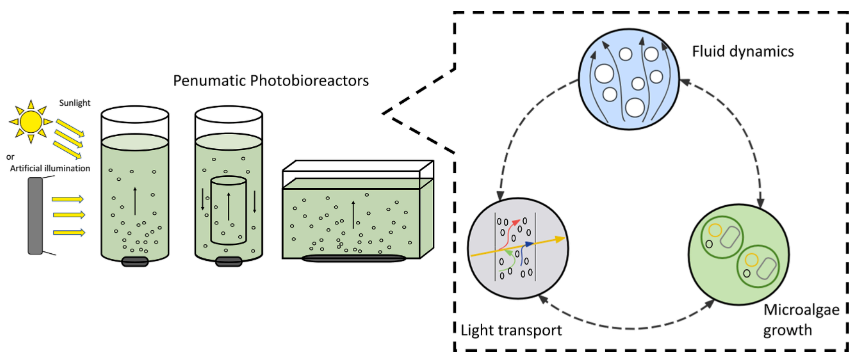
HIGHLIGHTS

- A multiphysics model was developed to predict the performance of microalgal pneumatic photobioreactors.
- The model fully incorporates multiphase fluid dynamics, light transport, and microalgae growth kinetics.
- The impact of bubble scattering on light transport was investigated.
- The effectiveness of the model was demonstrated through the simulation of an industrial scale outdoor photobioreactor.

GRAPHICAL ABSTRACT



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ABSTRACT

A novel and comprehensive multiphysics numerical simulation model for microalgal pneumatic photobioreactors was developed using the open-source Computational Fluid Dynamics (CFD) software OpenFOAM. This model integrates three distinct submodels: multiphase fluid dynamics, light transport, and microalgae growth. In particular, the light transport submodel accounts for light scattering caused by air bubbles. The model effectively addressed the challenges associated with the disparate time scales of the submodels, and its accuracy was validated by comparing with several experiments in different configurations. This study focuses on how an outdoor flat plate photobioreactor can maximize solar energy utilization. The results indicate that the daily productivity of the light-intensity adaptive reactor increased by 12.31 % to 26.31 % compared to those at constant biomass concentrations, demonstrating significant potential for employing this model in conjunction with automated systems to enhance productivity.

1. Introduction

Biological capture of CO_2 by photosynthetic microalgae has received widespread attention in recent years as more and more countries pursue carbon neutrality. In addition to CO_2 absorption, artificial cultivation of microalgae presents potential for large-scale applications in producing food, medicines, biofuels, and other value-added products (Adeniyi et al., 2018; Mayfield, 2020). Consequently, it is necessary to improve the design and performance of microalgal photobioreactors (PBRs) to

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Nomenclature

A	parameter in drag model, dimensionless
a_M	absorption coefficient of microorganism, $L\,m^{-1}\,g^{-1}$
C	global biomass concentration, $g\,L^{-1}$
C_{TD}	turbulence dispersion coefficient, dimensionless
C_{VM}	virtual mass coefficient, dimensionless
C_w	wall lubrication coefficient, dimensionless
c	local biomass concentration, $g\,L^{-1}$
D_e	effective diffusion coefficient, $m^2\,s^{-1}$
Me	kinetic parameter in PSU model, s^{-1}
Sc_t	turbulent Schmidt number, dimensionless
α	kinetic parameter in PSU model, $m^2\,\mu mol^{-1}$
β	kinetic parameter in PSU model, $m^2\,\mu mol^{-1}$
γ	kinetic parameter in PSU model, s^{-1}
δ	kinetic parameter in PSU model, s^{-1}
κ	kinetic parameter in PSU model, dimensionless
μ	local specific growth rate, s^{-1}
μ_t	turbulent viscosity, Pa s
σ_M	scattering coefficient of microorganism, $m^{-1}\,(g/L)^{-1}$
τ_{eff}	effective shear stress, $N\,m^{-2}$
Φ	scattering phase function
Ω	solid angle, degree

increase biomass yield and conversion while reducing energy consumption. Simulation of complex fluid within PBRs using CFD techniques can provide a cost-effective approach to improving reactor efficiency (Bitog et al., 2011; Ranganathan et al., 2022).

Accurately predicting microalgae growth in CFD simulations is critical but difficult due to the complex relationships between fluid dynamics, light transport, and microalgae growth. Fig. 1 highlights these relationships: 1) fluid motion facilitates the nutrient transfer and microalgae movement; 2) light intensity directly influences the photosynthetic rate of microalgae; 3) microalgae growth inversely affects the flow behavior and light transport; 4) in a pneumatic PBR, rising gas bubbles scatter light in all directions. Consequently, quantitative investigation of PBRs through CFD simulations is challenging because it requires not only the suitable individual submodels for fluid dynamics, light transport, and microalgae growth, but also an efficient framework to couple these submodels.

Various multiphase CFD models for PBR fluid dynamics have been developed. Particularly, the Eulerian approach, which is commonly used for large-scale cultivation, solves a multiphase model for both the culture medium and the gas bubbles to avoid tracking thousands of individual microalgae cells compared with the Lagrangian approach (Gao et al., 2018; 2017; Wang et al., 2021; Xu et al., 2020). Regarding light transport in PBRs, the Discrete Ordinates (DO) model allows for multi-dimensional and multi-wavelength light transport, offering advantages over the Beer-Lambert law and the two-flux approximation (Kong and Vigil, 2014). In terms of microalgae growth, the photosynthetic unit (PSU) model is introduced to account for temporal variations in light, including shorter light/dark cycles, which provides the benefit of being easily incorporated into fluid dynamics and light transport simulations (Allen and Holmes, 1986; Lavorel and Joliot, 1972; Stirbet et al., 2020).

Our previous studies did not account for the bubble scattering in pneumatic PBRs. Furthermore, the photon flux was represented by an algebraic expression instead of directly integrating it into microalgae growth, which limits its general applicability across different types and sizes of PBRs. Additionally, coupling these submodels presents challenges due to their various time scales: light transport is nearly instantaneous, fluid dynamics reach a steady state within seconds, and microalgae growth occurs over days. It is impractical to perform long-term fluid dynamics simulations and update the light distribution at each iteration.

In this work, we developed a comprehensive multiphysics numerical simulation model using OpenFOAM, which integrates a multiphase Eulerian model for fluid dynamics, a radiation model that can accommodate the bubble scattering effect, and a PSU-based kinetics model for microalgae growth. The framework effectively addresses challenges posed by disparate time scales during simulation, offering more accurate predictions for PBR performance. Section 2 details the governing equations of these three submodels and the solution algorithm. Section 3 outlines three different PBR configurations used for validation and application in this study, along with their respective simulation procedures. Section 4 describes and discusses the comparative results between simulations and experimental data and emphasizes the great

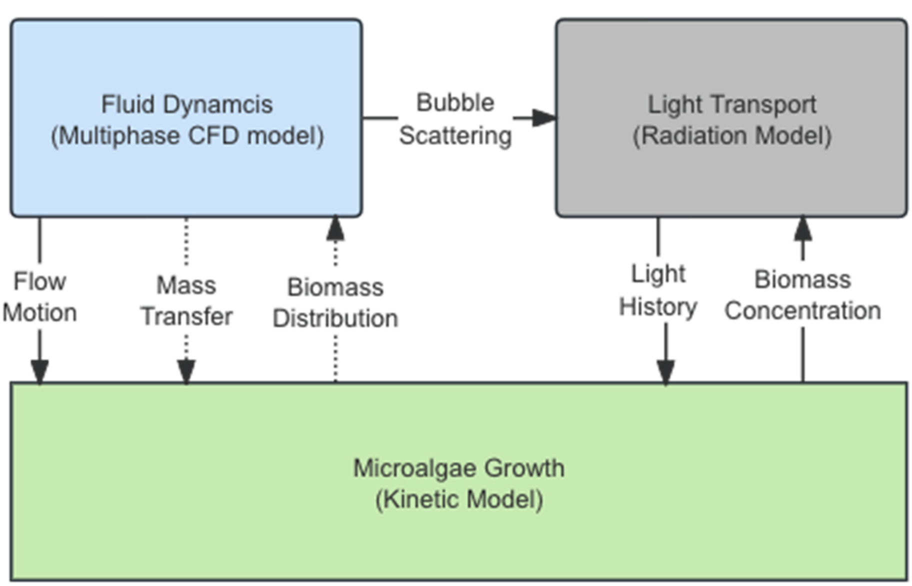

Fig. 1. Complex interaction between fluid dynamics, light transport, and microalgae growth kinetics.

potential of this model when combined with automated equipment to enhance productivity.

2. Materials and methods

2.1. Model description

2.1.1. Multiphase fluid dynamics

Microalgae cells drift with the liquid flow due to their low Stokes number. They are not treated as a separative solid phase in the multiphase fluid dynamics simulation but rather as passive concentration species in the liquid phase. Therefore, a two-fluid Eulerian approach was employed to study the liquid and gas flow dynamics:

$$\frac{\partial}{\partial t}(\alpha_i\rho_i)+\nabla\bullet\left(\alpha_i\rho_i\overrightarrow{U}_i\right)=0 \tag{1}$$

$$\frac{\partial}{\partial t}\left(\alpha_i\rho_i\overrightarrow{U}_i\right)+\nabla\bullet\left(\alpha_i\left(\rho_i\overrightarrow{U}_i\overrightarrow{U}_i\right)\right)=-\alpha_i\nabla p_i+\alpha_i\rho_i\overrightarrow{g}+\nabla\bullet\left(\alpha_i(\mu_i+\mu_i^T)\left(\nabla\overrightarrow{U}_i+\left(\nabla\overrightarrow{U}_i\right)^T\right)\right)+\overrightarrow{F}_i \tag{2}$$

where α_i, ρ_i, and $\overrightarrow{U}_i$ are the volume fraction, density, and velocity vector of phase i, respectively. p is the pressure, $\overrightarrow{g}$ is the gravity vector, μ and μ^T are the molecular viscosity and the dynamic viscosity due to turbulence, which was simulated using the mixture $k-\epsilon$ turbulence model introduced by Behzadi et al. (2004) and Lahey (2005). The interphase force $\overrightarrow{F}_i$, accounts for the interphase momentum transfer between gas and liquid, representing drag, lift, virtual mass, wall lubrication, and turbulence dispersion forces (see Supplementary Material).

2.1.2. Light transport

Based on the radiative transport theory, a photon beam traveling along the path loses its energy by absorption and out-scattering into the medium and gains energy by emission and in-scattering from the surroundings. In a heterogeneous medium such as pneumatic PBRs, rising gas bubbles can affect light scattering. Therefore, the so-called multiphase radiative transport equation (RTE) is solved:

$$\frac{dI_\lambda(\overrightarrow{r},\overrightarrow{s})}{ds}+K_\lambda I_\lambda(\overrightarrow{r},\overrightarrow{s})=\frac{\sigma_{L,\lambda,s}}{4\pi}\int_0^{4\pi}I_\lambda(\overrightarrow{r},\overrightarrow{s})\Phi_{L,\lambda}(\overrightarrow{s}',\overrightarrow{s})d\Omega+\frac{\sigma_{G,\lambda,s}}{4\pi}\int_0^{4\pi}I_\lambda(\overrightarrow{r},\overrightarrow{s})\Phi_{G,\lambda}(\overrightarrow{s}',\overrightarrow{s})d\Omega \tag{3}$$

where $I_\lambda(\overrightarrow{r},\overrightarrow{s})$ indicates the wavelength (λ) light intensity in direction $\overrightarrow{r}$ and location $\overrightarrow{s}$. $\Phi(\overrightarrow{s}',\overrightarrow{s})$ is the scattering phase function and Ω is the solid angle. The extinction coefficient K_λ can be expressed as the sum of light absorption and scattering by microalgae and light scattering by gas bubbles:

$$K_\lambda=(a_{M,\lambda}+\sigma_{M,\lambda,s})C(1-\alpha_G)+\sigma_{G,\lambda,s}\alpha_G \tag{4}$$

where $a_{M,\lambda}$ and $\sigma_{M,\lambda,s}$ are the absorption and scattering coefficients of microalgae, C is the global biomass concentration and α_G is the gas phase void fraction calculated in the multiphase fluid dynamics simulation.

The DO method is applied for solving the RTE, which is beneficial in coupling with fluid dynamics. The optical properties of microalgae are measured using laser diffraction and the Mie solution of Maxwell's equations for electromagnetic radiation interacting with a sphere. The responding details of the DO model and Mie's solution are shown in our previous research (Kong and Vigil, 2014).

Based on the fluid dynamics simulation, the bubbles can be assumed to be perfectly smooth spheres, leading to an "optically smooth" gas–liquid interface, resulting in specular reflection. Since the bubble size is much larger than the wavelength of the incident radiation, light scattering analysis may be based on the laws of reflection by large specularly reflecting spheres, as indicated by the "hemispherical reflectivity" concluded by Siegel and Howell (1992) (see Supplementary Material).

2.1.3. Microalgae growth

Microalgae grow photosynthetically, with light intensity being a crucial limiting factor. A PSU model refers to a cellular component responsible for photosynthetic processes, it assumes that a fraction of light-absorbing reaction centers exists in one of the three states: resting (x_1), active (x_2), and inhibited (x_3). Since these three states are considered components of the liquid phase, the local biomass concentration (c_t) and the concentration of resting and active state microorganisms (c_1, c_2) can be described by the species transport equation:

$$\frac{\partial}{\partial t}(\alpha_l c_t)+\nabla\bullet\left(\alpha_l\overrightarrow{U}_l c_t\right)=\nabla\bullet(\alpha_l D_e\nabla c_t)+S_t \tag{5}$$

$$\frac{\partial}{\partial t}(\alpha_l c_1)+\nabla\bullet\left(\alpha_l\overrightarrow{U}_l c_1\right)=\nabla\bullet(\alpha_l D_e\nabla c_1)+S_1 \tag{6}$$

$$\frac{\partial}{\partial t}(\alpha_l c_2)+\nabla\bullet\left(\alpha_l\overrightarrow{U}_l c_2\right)=\nabla\bullet(\alpha_l D_e\nabla c_2)+S_2 \tag{7}$$

$$c_1+c_2+c_3=c_t \tag{8}$$

where S_t is the source term based on microalgae growth, S_1 and S_2 are the source terms derived from the PSU model. D_e represents the effective turbulent diffusivity. T he local specific growth rate μ is computed as follows:

$$\mu=\kappa\gamma x_2-Me=\kappa\gamma\frac{c_2}{c_t}-Me \tag{9}$$

where κ is the yield of photosynthesis production to the transition from the active state to the resting state and Me is the maintenance coefficient that accounts for the effect of shear stress on the growth rate. A PSU model revised by Wu and Merchuk (2001) was used in this work (see Supplementary Material)

2.1.4. Solution algorithm

The multiphysics numerical simulation model for microalgal pneumatic photobioreactors was implemented based on the submodels described above. This model addresses several interrelated processes, each operating on vastly different time scales, which presents a significant challenge in coupling these submodels. In multiphase fluid dynamics, gas–liquid flows exhibit rapid fluctuations that require very small time steps for convergence, typically on the order of 10^{-4}s. Light transport is treated as time-independent in steady-state simulations due to the high speed of light. In contrast, microalgae growth is a considerably slower process that occurs over several days.

To address this challenge, the simulation framework employs an iterative approach to solve these processes (Fig. 2). Initially, the fluid dynamics are solved and the key flow parameters, such as time-averaged gas void fraction and liquid velocity, are recorded once a steady state is

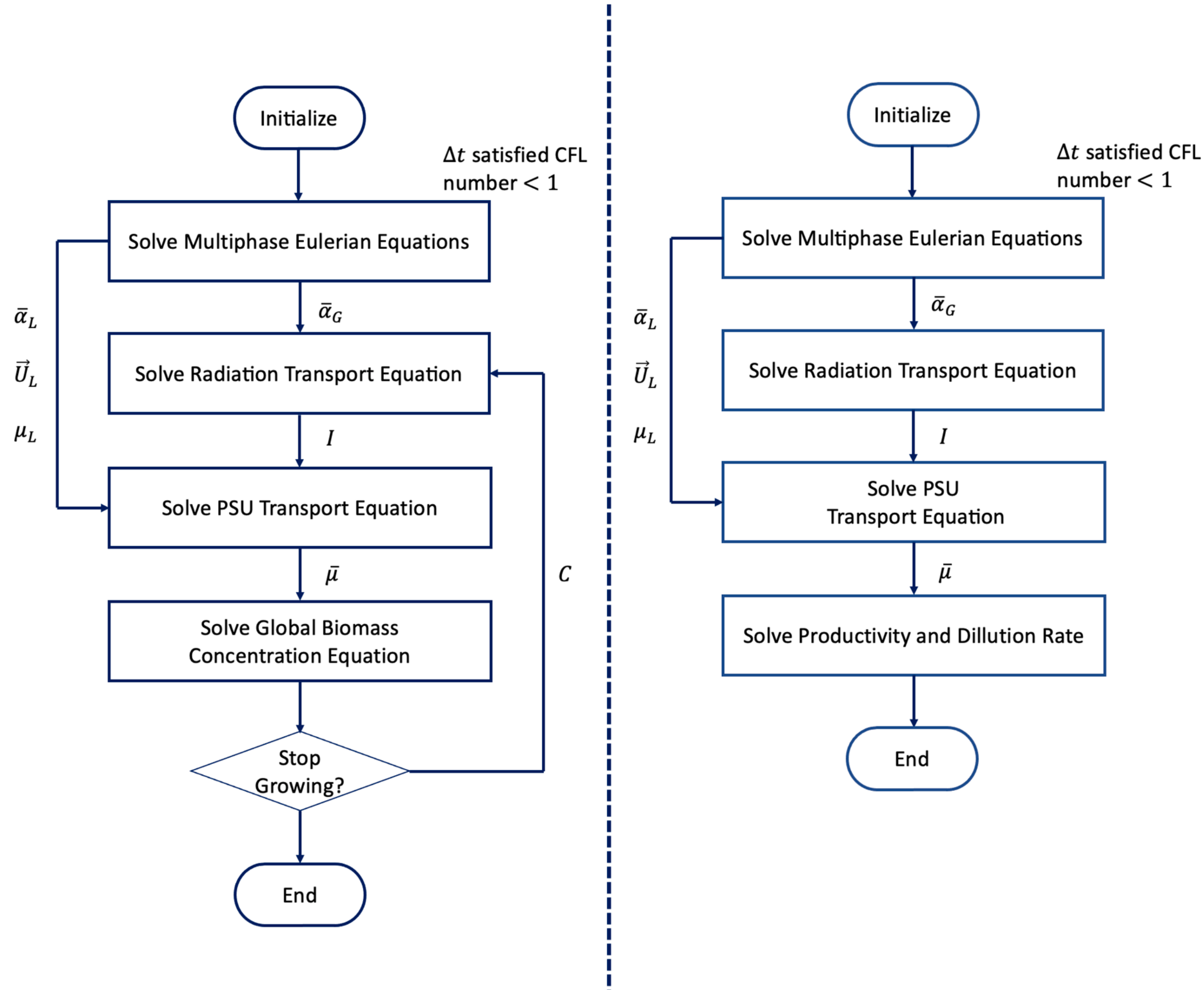

Fig. 2. Computational schemes of the comprehensive multiphysics numerical simulation model for batch reactor (left) and continuous reactor (right).

achieved. Due to the assumption that the periodic fluctuations in the flow have a negligible effect on light transport, these time-averaged quantities are subsequently used to simulate the steady-state light transport. Following this, microalgae growth is computed based on the results of the fluid dynamics and light distribution simulations. The cycle of recomputing light distribution, solving for growth, and updating biomass is repeated until biomass growth reaches saturation in the batch reactors. In case of continuous reactors, the process is similar, however, instead of updating biomass concentration, the simulation calculates global biomass productivity.

Furthermore, to effectively address the dynamics of prolonged biomass growth, the simulation adopts a stepwise approach. The Monod equation governs the temporal variations in biomass concentration:

$$\frac{dC}{dt} = P = \bar{\mu}C \tag{10}$$

where P is the productivity and $\bar{\mu}$ is the volume-averaged growth rate. A first-order exponential integrator method combined with an adaptive step size was employed to optimize computational efficiency by adjusting the time step based on the relative error between successive iterations. If the error between the calculated concentrations at consecutive time steps exceeds 1 %, the time step is reduced. Conversely, if the error is below 1 %, the time step is increased. This approach allows for large steps when biomass growth changes slowly and smaller steps when more precision is required, ensuring the solution remains numerically stable while minimizing unnecessary computational costs.

2.2. *CFD simulation*

Numerical CFD simulations were performed in OpenFOAM-v2206 to validate the model and to demonstrate its application in different types of PBR (Fig. 3). The simulations focused on three scenarios: light intensity validation, microalgae growth validation, and realistic outdoor PBR application.

2.2.1. *Light intensity validation*

Sun et al. (2016) conducted a comparative analysis of the light attenuation and biomass concentration for Chlorella vulgaris FACHB-31 (*C. vulgaris*) in a planar waveguide plate PBR (mPW-PBR) versus a bubble reactor (B-PBR), which are illustrated in Fig. 3(a). The gas phase consisted of air containing 5 % CO_2, which was supplied at a rate of 0.15 vol of air in volume of broth per minute via a gas sparger. 2D simulations employed uniform rectangular computational meshes consisting of 6,600 and 3,5000 cells for the mPW-PBR and B-PBR, respectively. The

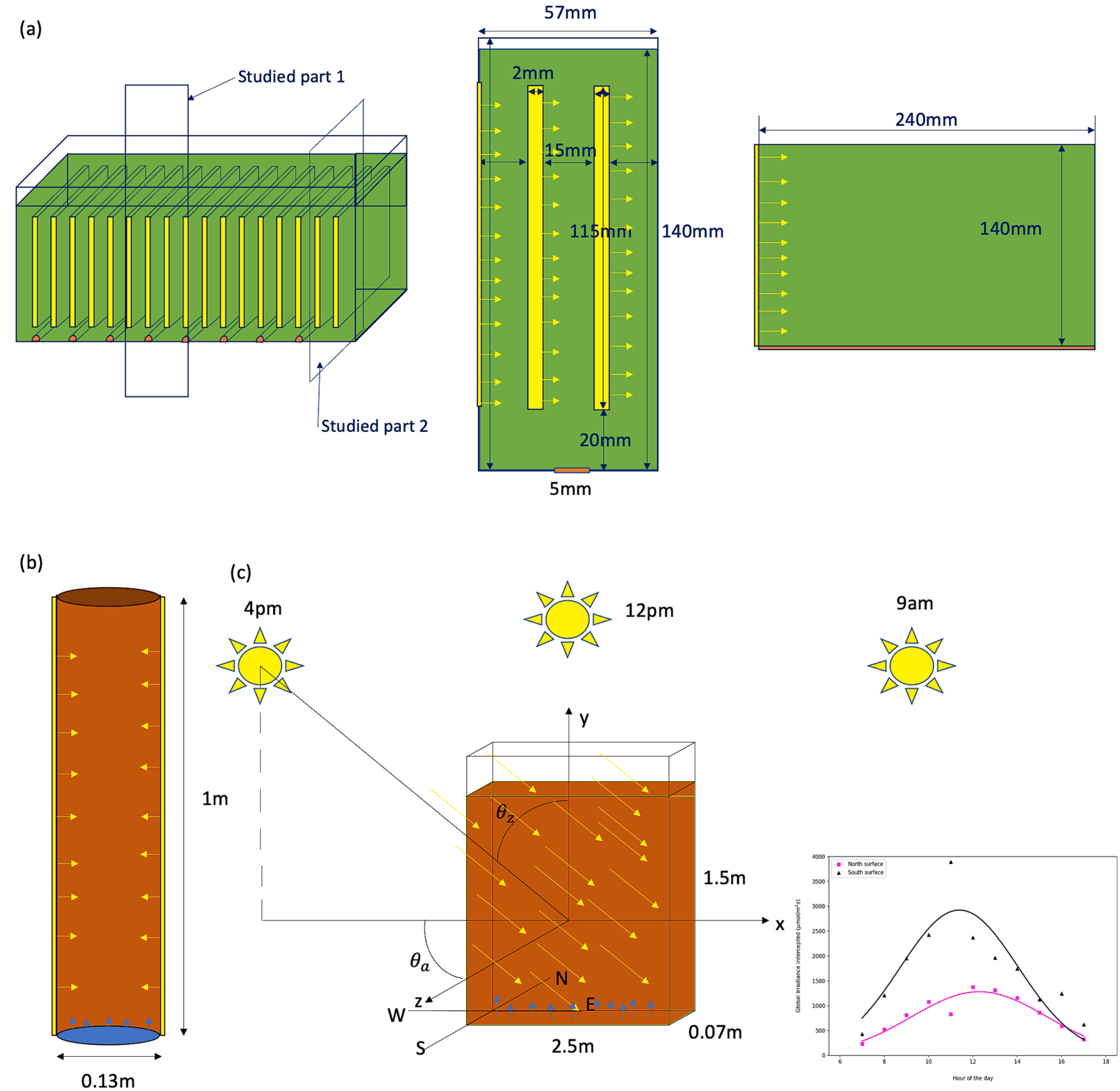

Fig. 3. Structural diagrams and configurations of three different PBRs in this study (a) the mPW-PBR and the studied part of mPW-PBR (middle), and B-PBR (right); (b) the bubble column reactor; (c) 250L flat plate photobioreactor (left) and light intensity interception by north and south surfaces during the whole day (right).

Table 1
Modeling constant and coefficients for *C. vulgaris* in light distribution simulation.

d_b	3×10^{-3}m	C_{w1}	-0.01
MW_g	29.75g mol^{-1}	C_{w2}	0.05
MW_l	18g mol^{-1}	C_{TD}	0.01
μ_g	$1.81 \times 10^{-5}\text{kg m}^{-1}\text{ s}^{-1}$	a_M	$100\text{L g}^{-1}m^{-1}$
μ_l	$1.15 \times 10^{-4}\text{kg m}^{-1}\text{ s}^{-1}$	σ_M	$400\text{L g}^{-1}m^{-1}$
A	24	ρ_λ	0.02
C_{VM}	0.25		

relevant modeling constants and coefficients are presented in Table 1. The mPW-PBR used a flat waveguide light source (95 μmol/m^{2}s), while the B-PBR had an external white LED light source (900 μmol/m^{2}s). Due to the absence of key parameters in the PSU model of *C. vulgaris* in biomass growth simulations, this section concentrated on validating the radiation model.

2.2.2. Microalgae growth validation

Wu and Merchuk (2002) conducted a study on the *Porphyridium* sp. cultivation in a bubble-column PBR (Fig. 3(b)), providing reliable parameters for PSU modeling of *Prophyridium* sp. They also employed a circulation time approach to simulate the movement trajectories of microalgae, integrating this approach with the PSU model to predict microalgae growth. In this study, the reliability of our multiphysics numerical simulation model was validated by comparing biomass growth predictions. The fluid dynamics parameters, optical properties, and kinetics parameters in the PSU model for *Porphyridium sp.* are

Table 2
Modeling constant and coefficients for *Prophyridium* sp. in microalgae growth simulations.

d_b	3×10^{-3}m	α	$0.001935\,m^2\,(\mu E)^{-1}$
MW_g	$29.75g\,mol^{-1}$	β	$5.7848 \times 10^{-7}\,m^2(\mu E)^{-1}$
MW_l	$18g\,mol^{-1}$	γ	$0.1460\,s^{-1}$
μ_g	$1.81 \times 10^{-5}kg\,m^{-1}\,s^{-1}$	δ	$0.0004796\,s^{-1}$
μ_l	$1.15 \times 10^{-4}kg\,m^{-1}\,s^{-1}$	κ	0.0003647
A	24	Me_0	$1.64 \times 10^{-5}s^{-1}$
C_{VM}	0.25	$D_{L,l}$	5.5×10^{-10}
C_{w1}	−0.01	k_m	$7.8 \times 10^{-3}Pa^{-1}$
C_{w2}	0.05	τ_c	1Pa
C_{TD}	0.01	Sc_t	1.1
a_M	$30L\,g^{-1}m^{-1}$	ρ_λ	0.472
σ_M	$80L\,g^{-1}m^{-1}$		

presented in Table 2. A 3D simulation with 112,500 cells was conducted for three different superficial gas velocities (0.16, 0.32, 0.82 cm/s) under a constant light source at the reactor's surface (250 $\mu molm^{-2}s^{-1}$).

The dynamic processes of biomass growth predicted by this model will be compared with the results of experiments and simulations using the circulation time approach.

2.2.3. Outdoor reactor simulation

To demonstrate the industrial application of this simulation model, a simulation was performed for an outdoor continuous PBR located in Shantou, China (23°20′50″N, 116°38′35″E) on a specific day: the fall equinox (September 22nd). Solar radiation data were obtained from the National Solar Radiation Database (NSRD). The flat plate reactor is considered to have the highest productivity among pneumatic PBRs due to its large light-exposed surface area-to-volume ratio (Aslanbay Guler et al., 2019). In addition to improving mixing efficiency by incorporating mixers or baffles within the PBR, further research should investigate methods to optimize sunlight utilization for improved reactor performance. A vertically placed 250L flat plate reactor facing south for cultivating *Porphyridium* sp. was modeled using 60,000 grid cells and the same parameters as presented in Table 2 (Sierra et al., 2008). The

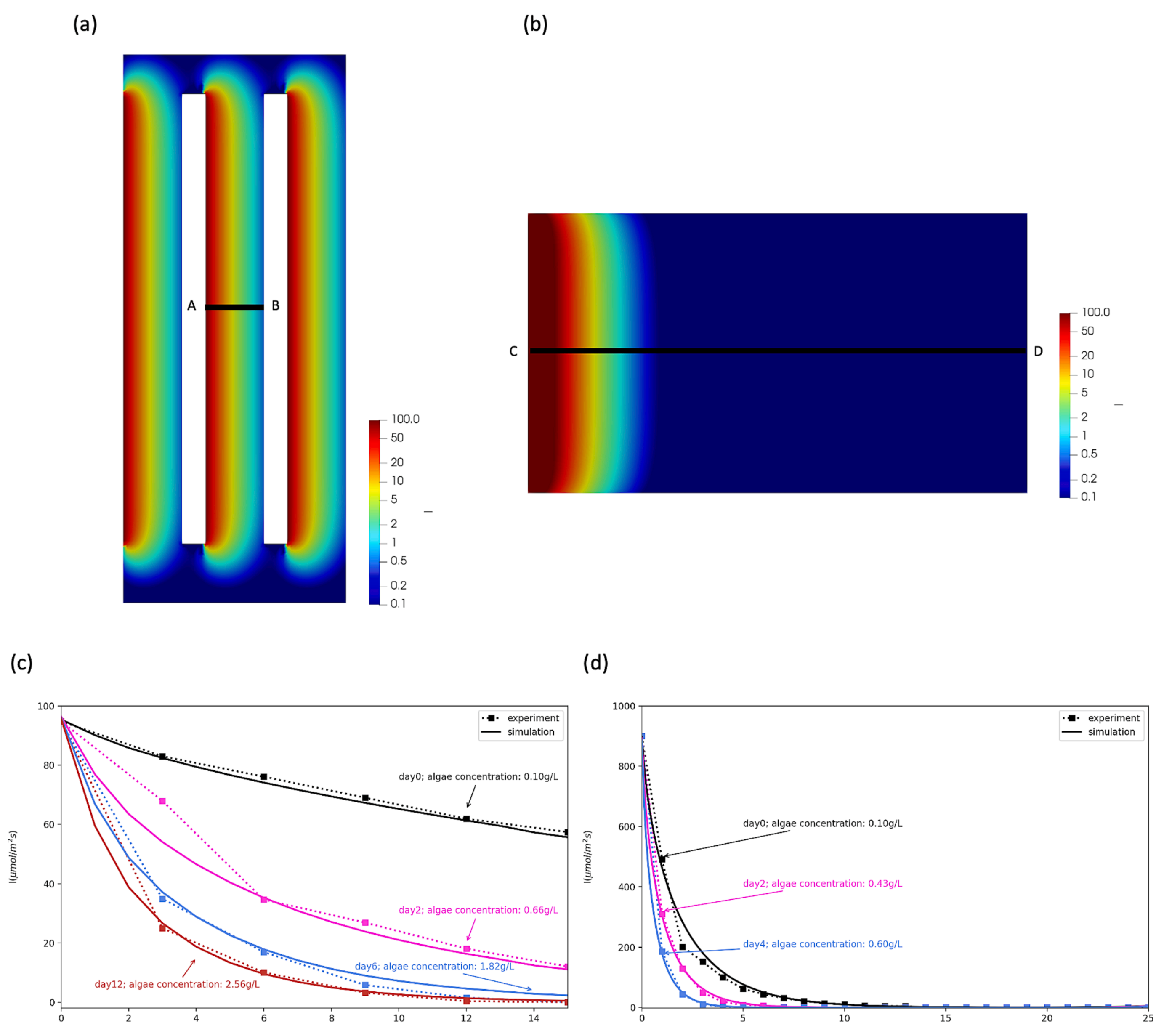

Fig. 4. Light distribution profiles at microalgae concentration C = 0.6 g/L in (a) mPW-PBR, and (b) B-PBR; Comparison of simulation predictions with corresponding experimental measurements for the light intensity along the line segments (c) AB in mPW-PBR, and (d) CD in B-PBR.

superficial gas velocity is set to $U_g = 0.76$ cm/s. The two largest surfaces (facing south and north) are assumed to be light-receiving surfaces, while the remaining surfaces are treated as reflecting surfaces. Fig. 3(c) shows the configuration of the flat plate PBR and the light intensity interception by these two surfaces at different times throughout the day (see Supplementary Material). Microalgae growth productivities were assessed for nine different concentrations at a specific time. Based on the database of the concentration-productivity relationship at that time, the polynomial fitting method was used to determine the optimal cultivation concentration and maximum productivity. The corresponding dilution rate was then calculated based on the maximum productivity. By incorporating weather forecast information, automated equipment can be utilized in conjunction with this model to adjust the liquid flow and biomass concentration at different times, achieving maximum productivity.

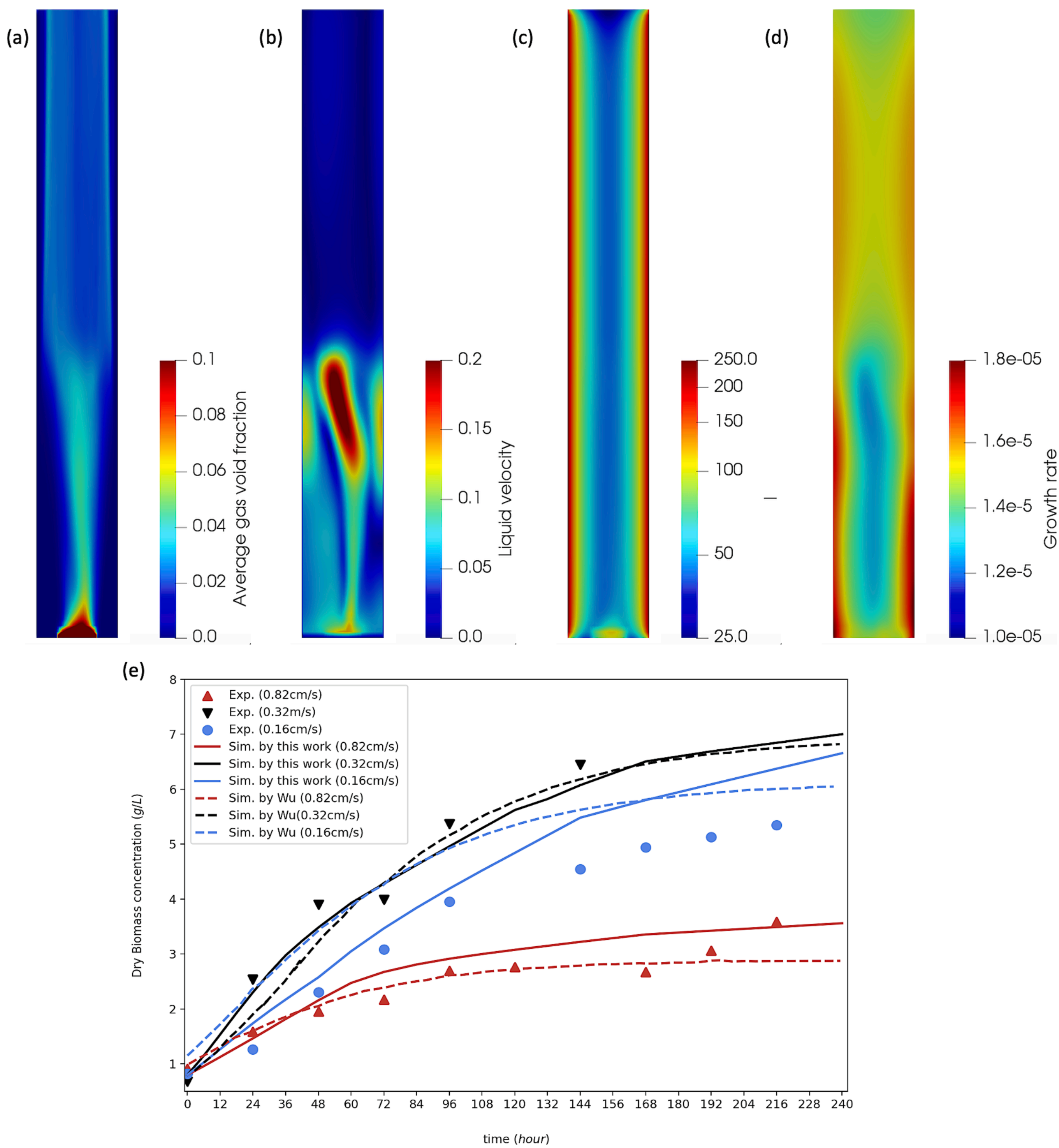

Fig. 5. The sample profiles of x-y section (a) average gas volume fraction, (b) liquid velocity, (c) light distribution, (f) biomass growth rate; and (e) comparison of microalgae growth curve predictions with experiments.

3. Results and discussion

3.1. Light intensity simulation results

Fig. 4(a) and (b) show the light distribution for the mPW-PBR and B-PBR at 0.6 g/L biomass concentration. The mPW-PBR exhibits better light exposure across the reactor, with minimal light dead zones. Quantitative comparisons between the simulation and experiment light intensities along the optical path in the center of the reactor (lines AB and CD in Fig. 4(a) and (b)) at different biomass concentrations, as depicted in Fig. 4(c) and (d), validate the accuracy of the model's prediction regarding light attenuation in both reactors.

Light intensity decreases with increasing biomass due to absorption and with distance from the light source. As shown in Fig. 4(a) and (c), the short light path in the mPW-PBR facilitates light distribution. In contrast, the B-PBR (Fig. 4(b) and (d)) experiences inadequate light distribution, which directly leads to a slower growth rate.

3.2. Microalgae growth simulation results

Fig. 5(a)–(d) illustrates the correlation between biomass growth rate and factors such as average gas void fraction, liquid velocity, and light distribution at a superficial gas velocity $U_g = 0.82\,cm/s$. Biomass growth decreases in areas with higher gas void fractions, higher liquid flow, and lower light intensity, resulting in a heterogeneous distribution of the biomass growth rate.

Both the experimental data and the simulation results are depicted in Fig. 5(e). The results indicate similar trends of microalgae growth at different superficial gas velocities. An appropriate increase in gas flow rate enhances biomass growth, while excessive gas flow rates generate high shear forces that can inhibit cell growth or even damage cells. The model effectively captures the complex relationship between biomass accumulation and gas flow rate and. In comparison to the simulation based on the circulation time approach, the results demonstrate that, particularly during the early stages of cultivation (0–4 days), the model developed in this study offers greater accuracy in predicting biomass growth.

3.3. Outdoor reactor simulation results

Light intensity attenuation at typical times (9 a.m., noon, and 4p.m.) for a biomass concentration of 1.0 g/L is depicted in Fig. 6(a). The dots represent the light intensity on the outer surface of the glass. Significant drops in light intensity exist at the glass-air interface and the glass-broth interface due to strong normal light reflection. The optimal biomass concentrations and maximum productivities at these typical times are illustrated in Fig. 6(b), which vary significantly with changing light conditions throughout the day. Therefore, adjusting the biomass concentration to specific levels throughout the day can greatly enhance productivity. The daily variation curve of the optimal biomass concentration and the dilution rate are summarized in Fig. 6(c). The superficial liquid velocity calculated from the dilution rat is on the order of 10^{-5}m/s, which is much smaller than the set superficial gas velocity. Consequently, the effect of liquid phase inlet and outlet flow on fluid dynamic performance can be neglected.

When cultivating with a constant concentration, light inhibition is likely to occur during the morning and afternoon due to insufficient solar radiation. In contrast, the strong solar radiation at noon can easily lead to over-saturation (Krichen et al., 2021). Thus, it is necessary to design an automated light-adaptive PBR to maximize the utilization of

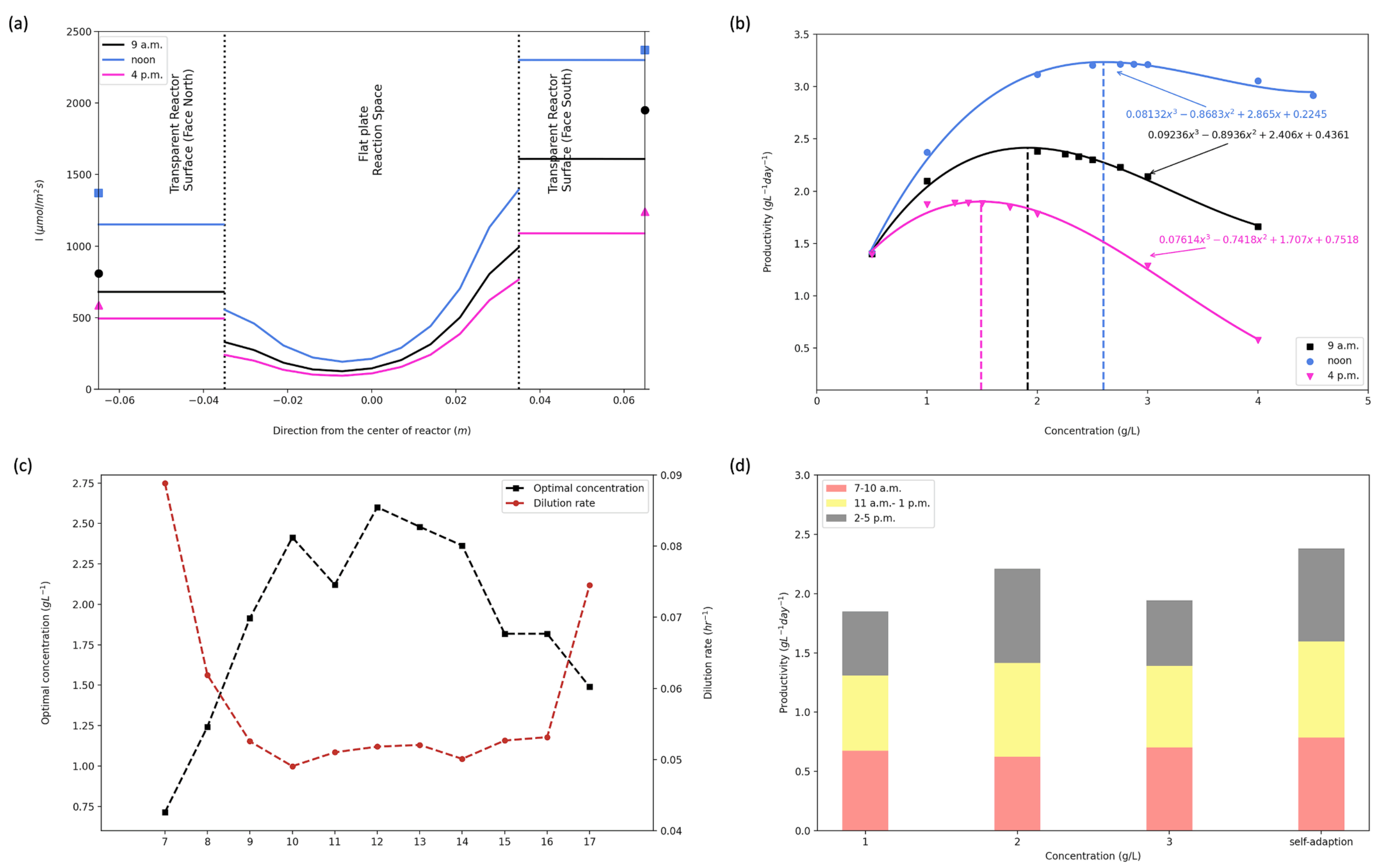

Fig. 6. (a) Average light intensity attenuation along the light path under three light conditions; (b) The productivity as a function of the biomass concentration under three different light conditions; (c) Optimum biomass concentration and dilution rate at different hours of the day; (d) The daily productivity at biomass concentration with light intensity adaptation taken and at constant biomass concentrations.

solar energy throughout the day. Fig. 6(d) compares the daily productivity of an automated light-adaptive PBR to a traditional PBR. The results indicate that the automated light-adaptive PBR can achieve an increase in productivity (2.38 g/L/day) from 12.31 % to 26.31 % compared to the traditional system with constant biomass concentrations (1.0, 2.0, and 3.0 g/L).

The development of an automated light-adaptive PBR presents a significant advancement in optimizing microalgae productivity under fluctuating natural light conditions. By continuously mirroring the real-time light intensity variations and adjusting biomass concentration accordingly, this digital twin model minimizes periods of light inhibition during lower light hours and prevents over-saturation at peak sunlight, ensuring optimal photosynthetic activity and resource use throughout the day.

4. Conclusion

This study presents a comprehensive numerical simulation model for analyzing microalgal pneumatic photobioreactors. The model integrates three critical submodels: multiphase fluid dynamics, light transport, and microalgae growth. It offers a novel approach to addressing the challenge of disparate time scales through iterative coupling. A key innovation of this model is its ability to account for light scattering caused by gas bubbles.

Validation of the model against experimental data confirmed its reliability. Furthermore, the model was applied to an industrial-scale outdoor photobioreactor, demonstrating its ability to optimize productivity under different radiation conditions by integrating with the automated control system.

CRediT authorship contribution statement

Ruofan Wang: Writing – original draft, Visualization, Validation, Software. **Xiaobo Yao:** Formal analysis, Data curation. **Raphael Semiat:** Writing – review & editing, Resources. **Bo Kong:** Writing – review & editing, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.131962.

Data availability

Data will be made available on request.

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